Total No. of Printed Pages:02

SUBJECT CODE NO: - RNP_8008
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - I and Sem - II
Examination November / December 2025 (To be held in February-2026)
Engineering Physics

[Time: 3 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no. 1 is compulsory
3) Solve any four questions from Q. nos 2, 3, 4, 5, 6 and 7

Section-A

Q1 **Solve the following questions (any 10)** 20

a) Newton's rings are formed due to:
A) Diffraction **B)** Polarization **C)** Interference **D)** Reflection

b) Type-II superconductors are also known as:
A) Hard superconductors **B)** Soft superconductors **C)** Perfect conductors **D)** Magnetic insulators

c) Michelson interferometer works on the principle of:
A) Diffraction **B)** Interference **C)** Refraction **D)** Photoelectric effect

d) Ultrasonic waves have frequencies...........

e) Bragg's law is given by............

f) What is interference of light?

g) Define De-Broglie wavelength

h) Write any two applications of X-ray diffraction (XRD).

i) What is quantum tunneling

j) Define single-mode optical fiber

k) A He-Ne laser emits light of frequency4.74×10^14 Hz. Calculate the wavelength.

l) The energy level of electrons at absolute zero is called ___________.

Section-B

Q2 A. Explain the formation of Newton's rings and derive an expression for the radius of bright and dark rings. 05

B. Write a short notes on: i) Electromagnetic waves ii) Electromagnetic spectrum 05

Q3 A. Explain the production and detection of X-rays. Discuss continuous and characteristic spectra. 05

B. Explain Davisson-Germer experiment 05

Q4	A.	Explain superconductivity, Meissner effect and discuss Type-II superconductors	05
	B.	Explain structure of optical fiber with suitable diagram, refractive index of core 1.48 and that of cladding is 1.47 in an optical fiber calculate numerical aperture	05
Q5	A.	Describe the construction, working, and applications of the Michelson interferometer.	05
	B.	Explain the Photoelectric effect with diagram.	05
Q6	A.	Derive Bragg's law and explain its applications in X-ray diffraction and lattice parameter determination	05
	B.	The spacing between principal planes of crystal is 2.82 $\dot{A}$. The first order Bragg reflection occurs at an angle of 9^0. What is the wavelength of X-rays?	05
Q7	A.	The wavelength of the X-rays is 0.07which is diffracted by a plane of salt with 0.28 nm. As the lattice constant. Determine the glancing angle for the second order diffraction. Assume the value of the salt flame to be 110 and the given salt is Rock Salt	05
	B.	Explain the ruby Laser?	05